Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 06: Dispersion 2

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Low-frequency behavior ($\omega \sim 0$)

- Im(ε)->∞, does this make sense?

High-frequency behavior ($\omega \to \infty$)

- In insulators ($\omega_j > 0$)

$$\frac{\varepsilon}{\varepsilon_0} = 1 + \frac{Ne^2}{\varepsilon_0 m} \sum_j (\omega_{0j}^2 - \omega^2 - i\omega\Gamma_j)^{-1}$$

$$= 1 - \frac{Ne^2}{\varepsilon_0 m} \sum_j \frac{1}{\omega^2} = 1 - \frac{Ne^2 Z}{\varepsilon_0 m \omega^2}$$

$Z = \sum_j 1$

Z is the number of electrons in one molecules.

$$\omega_p^2 = \frac{Ne^2 Z}{\varepsilon_0 m}$$

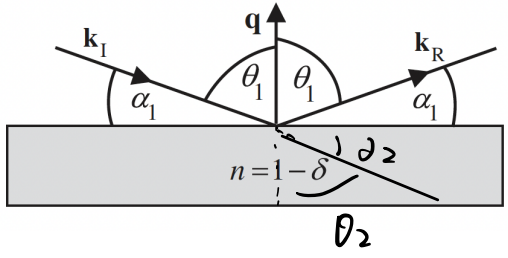
ω_p is called plasma frequency

$$\frac{\varepsilon}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2} < 1$$

Example: X-ray

- At X-ray frequency, most materials have an index of refraction less than 1.

Almost Total External Reflection A plane wave in vacuum with wave vector $\mathbf{k}_I$ reflects from a non-magnetic sample into a plane wave with wave vector $\mathbf{k}_R$. At X-ray wavelengths, the index of refraction of essentially all matter is very slightly less than 1. Typically, $n \approx 1 - \delta$ with $\delta \sim 10^{-5}$. This makes the phenomenon of total *external* reflection of X-rays possible. Let $\mathbf{q} = \mathbf{k}_R - \mathbf{k}_I$ and show that the Fresnel reflection coefficients for s- and p-polarization, R_s and R_p, both vary as q^{-4} when the angle $\alpha_1 \ll 1$ but not quite small enough for total external reflection to occur.



$\sin\theta_1 = n \sin\theta_2$

$\Rightarrow \cos\theta_1 = n \cos\theta_2$

$1 - \frac{1}{2}\theta_1^2 = n - \frac{n}{2}\theta_2^2$

$1 - n = \frac{1}{2}\theta_1^2 - \frac{n}{2}\theta_2^2$

$\delta = \frac{1}{2}\theta_1^2 - \frac{1}{2}\theta_2^2 + \frac{\delta}{2}\theta_2^2$

$\theta_1^2 - \theta_2^2 = 2\delta$

Example: X-ray

$$R_S = \left(\frac{\cos\theta_1 - n\cos\theta_2}{\cos\theta_1 + n\cos\theta_2} \right)^2 = \left(\frac{\sin\alpha_1 - n\sin\alpha_2}{\sin\alpha_1 + n\sin\alpha_2} \right)^2$$

$$= \left(\frac{\alpha_1 - \alpha_2}{\alpha_1 + \alpha_2} \right)^2 = \left(\frac{\alpha_1^2 - \alpha_2^2}{(\alpha_1 + \alpha_2)^2} \right)^2$$

$$= \left(\frac{2\delta}{4\alpha_1^2} \right)^2 \Rightarrow R_S \propto \alpha_1^{-4}$$

$$\vec{q} = \vec{k_R} - \vec{k_I} \qquad q = 2k\sin\alpha_1 \approx 2k\alpha_1$$

$$\Rightarrow \alpha_1 = \frac{q}{2k} \Rightarrow R_S \propto q^{-4}$$

Example: Plasma

Free electrons. $\omega_0 = 0$, $\gamma = 0$

$$\frac{\varepsilon}{\varepsilon_0} = 1 - \frac{Ne^2}{\varepsilon_0 m \omega^2} \qquad \omega_p^2 = \frac{Ne^2}{\varepsilon_0 m}$$

$$\frac{\varepsilon}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2}$$

$\omega < \omega_p$, $\frac{\varepsilon}{\varepsilon_0} < 0$

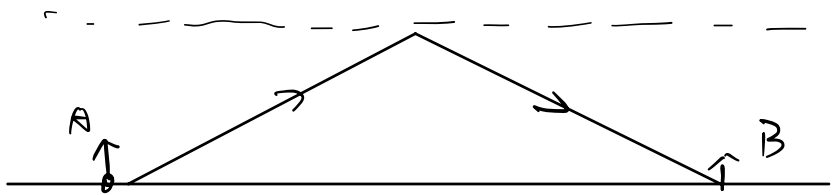
Example: Plasma

- Lab $N \sim 10^{18} \sim 10^{23}/m^3 \Rightarrow \omega_p = 6\times10^{10} \sim 6\times10^{12}$ Hz

  Microwaves.

- Upper atmosphere 100 km - 600 km

  $N \sim 10^{12}/m^3 \Rightarrow \omega_p = 60$ MHz Radio waves



Metal

- $\omega \ll \gamma_0$ $\varepsilon = \varepsilon_b + i\frac{Ne^2}{m\omega\gamma_0} = \varepsilon_b + i\frac{b}{\omega}$

$$b = \frac{Ne^2}{m\gamma_0}$$

- $\gamma_0 \ll \omega \ll \omega_p$ $\varepsilon = \varepsilon_b + i\frac{Ne^2}{m\omega(-i\omega)} = \varepsilon_b - \frac{Ne^2}{m\omega^2} = \varepsilon_b - \varepsilon_0\frac{\omega_p^2}{\omega^2}$

- $\omega \gg \omega_p$ $\varepsilon \approx \varepsilon_b$

Complex wave number

- ϵ is real, e.g., total internal reflection

$k = w\sqrt{\mu\varepsilon}$ μ is a real number

k_y, k_z

$\vec{k}\cdot\vec{k} = w^2\mu\varepsilon$ $\vec{k} = \vec{q} + i\vec{p}$

$(\vec{q}+i\vec{p})\cdot(\vec{q}+i\vec{p}) = w^2\mu\varepsilon$

$q^2 - p^2 + 2i\vec{p}\cdot\vec{q} = w^2\mu\varepsilon \Rightarrow \begin{cases} q^2 - p^2 = w^2\mu\varepsilon \\ \vec{p}\cdot\vec{q} = 0 \Rightarrow \vec{p}\perp\vec{q} \end{cases}$

Complex wave number

- ϵ is complex

$\vec{k}$ is in z direction.

$$k = \omega\sqrt{\mu\varepsilon}$$

$$k = \beta + i\frac{\alpha}{2}$$

$$e^{ikz} = e^{i\beta z}\, e^{-\frac{\alpha}{2}z}$$

$$\left(\beta + i\frac{\alpha}{2}\right)^2 = \omega^2 \mu\varepsilon$$

$$\Rightarrow \begin{cases} \beta^2 - \dfrac{\alpha^2}{4} = \dfrac{\omega^2}{c^2}\,\mathrm{Re}\dfrac{\varepsilon}{\varepsilon_0} \\ \beta\alpha = \dfrac{\omega^2}{c^2}\,\mathrm{Im}\dfrac{\varepsilon}{\varepsilon_0} \end{cases}$$

if $\alpha \ll \beta$, (no resonance)

$$\begin{cases} \alpha = \dfrac{\mathrm{Im}\,\varepsilon}{\mathrm{Re}\,\varepsilon}\beta \\ \beta = \dfrac{\omega}{c}\sqrt{\mathrm{Re}\dfrac{\varepsilon}{\varepsilon_0}} \end{cases}$$

$$E \propto e^{-\frac{\alpha}{2}z}$$

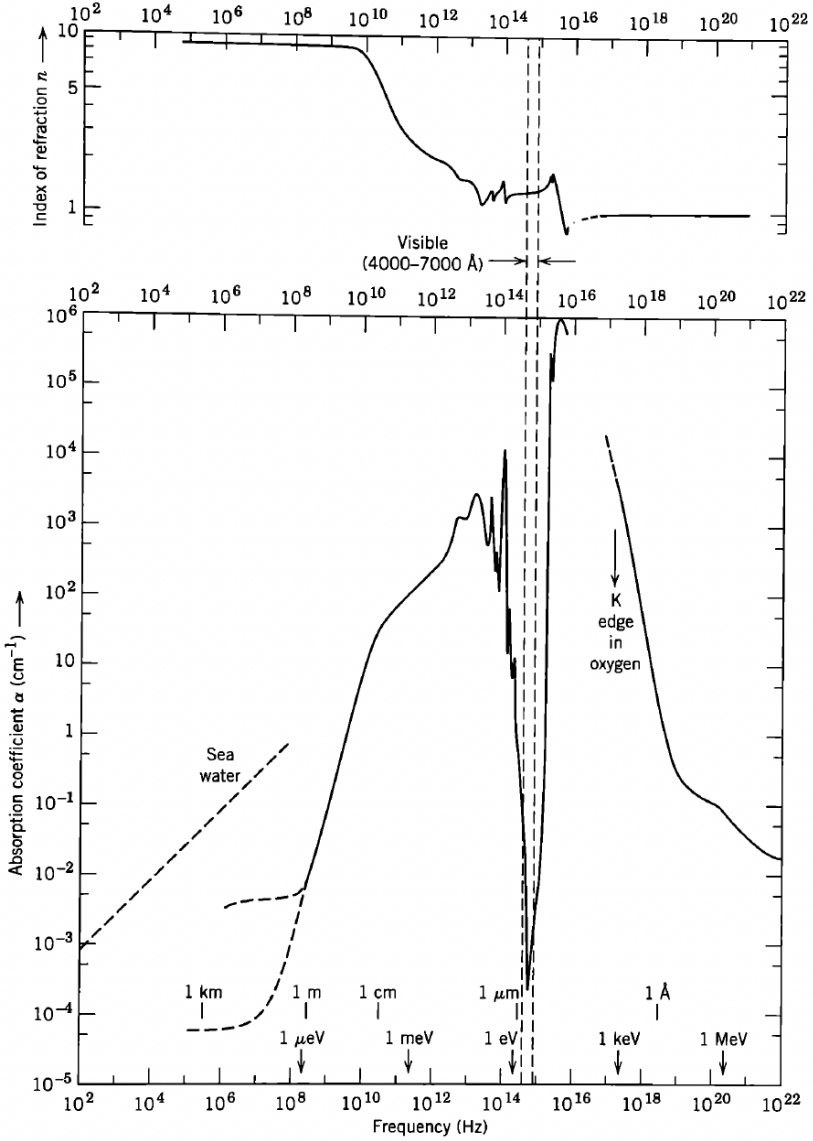
$$I \propto e^{-\alpha z}$$

Dispersion relation

$$k = \omega \sqrt{\mu \varepsilon(\omega)}$$

$$\omega = \omega(k)$$

Water


Index of refraction n
Visible
(4000–7000 Å)
Absorption coefficient α (cm⁻¹)
Sea
water
K
edge
in
oxygen
1 km
1 m
1 cm
1 μm
1 Å
1 μeV
1 meV
1 eV
1 keV
1 MeV
Frequency (Hz)